

Solar C appendix

Sun · C · downloadable appendix

Solar carbon: 8.430 ± 0.072

Asplund, Amarsi & Grevesse 2021: 8.46 ± 0.04 · this measurement -0.030 dex

Kitt Peak (1984, molecfit) · VIS · Reference Grade · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · $n = 3$.
The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

8.430 ± 0.072

± 0.052 stat · ± 0.049 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

red-optical

8.488 ± 0.069

± 0.036 stat · ± 0.059 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	17 products -- C I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	22 products -- C I 5052 Å (single line); C I 5380 Å (single line); C I · Codex-graded lines; CH G-band (molecular band); C ₂ Swan band (molecular band)
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS A(C) dex; axis 8.20-8.78

C I 5052 Å (single line) · 1D LTE Codex Grade · 1 line		8.444 ±0.089 stat ±0.103 syst
C I 5380 Å (single line) · 1D LTE Codex Grade · 1 line		8.528 ±0.087 stat ±0.218 syst
C ₂ Swan band (molecular band) · 1D LTE Codex Grade · 167 lines		8.419 ±0.010 stat ±0.186 syst
CH G-band (molecular band) · 1D LTE Codex Grade · 184 lines		8.509 ±0.006 stat ±0.089 syst
C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		8.483 ±0.019 stat ±0.109 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.463 ±0.019 stat ±0.113 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.476 ±0.019 stat ±0.113 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.476 ±0.031 stat ±0.075 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.488 ±0.033 stat ±0.075 syst

VIS — IAG FTS A(C) dex; axis 8.20-8.78

C I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		8.373 ±0.114 stat ±0.119 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.355 ±0.115 stat ±0.123 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.367 ±0.113 stat ±0.122 syst

VIS — Kitt Peak (Kurucz 2005) A(C) dex; axis 8.20-8.78

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines		8.467 ±0.059 stat ±0.048 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines		8.449 ±0.059 stat ±0.057 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines		8.460 ±0.059 stat ±0.057 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.363 ±0.086 stat ±0.095 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.375 ±0.085 stat ±0.095 syst

VIS — Kitt Peak (1984, molecfit) A(C) dex; axis 8.20-8.78

C I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 3 lines



8.450
 ± 0.052 stat ± 0.038 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 3 lines



8.430
 ± 0.052 stat ± 0.049 syst

C I · Asplund et al. 2021 lines · 1D NLTE
(Bergemann)
Reference Grade · 3 lines



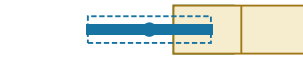
8.443
 ± 0.052 stat ± 0.049 syst

C I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 2 lines



8.376
 ± 0.085 stat ± 0.082 syst

C I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 2 lines



8.388
 ± 0.084 stat ± 0.082 syst

VIS axis: 8.20 · 8.35 · 8.49 · 8.63 · 8.78 A(C) dex

red-optical — IAG FTS A(C) dex; axis 8.39-8.76

C I · Asplund et al. 2021 lines · 1D LTE
Reference Grade · 10 lines



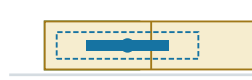
8.529
 ± 0.039 stat ± 0.052 syst

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 10 lines



8.488
 ± 0.036 stat ± 0.059 syst

C I · Asplund et al. 2021 lines · 1D NLTE
(Bergemann)
Reference Grade · 10 lines



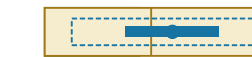
8.490
 ± 0.035 stat ± 0.060 syst

C I · Codex-graded lines · 1D LTE
Codex Grade · 6 lines



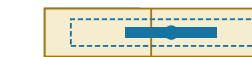
8.566
 ± 0.037 stat ± 0.080 syst

C I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 6 lines



8.528
 ± 0.040 stat ± 0.085 syst

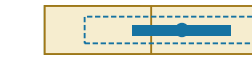
C I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 6 lines



8.527
 ± 0.039 stat ± 0.085 syst

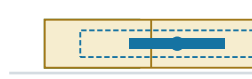
red-optical — Kitt Peak (Kurucz 2005) A(C) dex; axis 8.39-8.76

C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)
Reference Grade · 10 lines



8.536
 ± 0.042 stat ± 0.082 syst

C I · Asplund et al. 2021 lines · 1D NLTE
(Bergemann)
Reference Grade · 10 lines



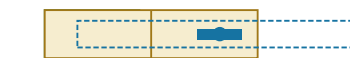
8.532
 ± 0.041 stat ± 0.082 syst

C I · Codex-graded lines · 1D LTE
Codex Grade · 6 lines



8.610
 ± 0.018 stat ± 0.115 syst

C I · Codex-graded lines · 3D NLTE (Amarsi)
Codex Grade · 6 lines



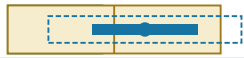
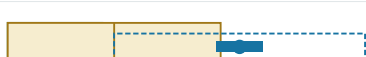
8.568
 ± 0.019 stat ± 0.120 syst

C I · Codex-graded lines · 1D NLTE (Bergemann)
Codex Grade · 6 lines



8.563
 ± 0.018 stat ± 0.119 syst

red-optical — Kitt Peak (1984, molecfit) A(C) dex; axis 8.39-8.76

C I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 10 lines		8.576 ±0.047 stat ±0.076 syst
C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 10 lines		8.536 ±0.045 stat ±0.082 syst
C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 10 lines		8.537 ±0.043 stat ±0.082 syst
C I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		8.616 ±0.020 stat ±0.106 syst
C I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 6 lines		8.579 ±0.022 stat ±0.111 syst
C I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 6 lines		8.582 ±0.022 stat ±0.110 syst

red-optical axis: 8.39 · 8.48 · 8.57 · 8.66 · 8.76 A(C) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every carbon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the carbon abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C₂, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(C)	Largest uncertainty terms (dex)
VIS	C I · Codex-graded lines	HARPS, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	2	8.355–8.476 (3D-NLTE)	continuum placement 0.05, line strengths (gf) 0.04
VIS	C I 5052 Å (single line)	HARPS	1D-LTE	1	8.444 (1D-LTE)	continuum placement 0.09, fit / line-to-line scatter 0.09
VIS	C I 5380 Å (single line)	HARPS	1D-LTE	1	8.528 (1D-LTE)	continuum placement 0.15, spread between spectra 0.12

Band	What was measured	Solar spectra	Treatments	Lines	Result A(C)	Largest uncertainty terms (dex)
VIS	C ₂ Swan band (molecular band)	HARPS	1D-LTE	167	8.419 (1D-LTE)	microturbulence 0.12, line strengths (gf) 0.08
VIS	CH G-band (molecular band)	HARPS	1D-LTE	184	8.509 (1D-LTE)	line strengths (gf) 0.06, spread between spectra 0.05
VIS	C I · Asplund et al. 2021 lines	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	3	8.430–8.463 (3D-NLTE)	fit / line-to-line scatter 0.05, line strengths (gf) 0.03
red-optical	C I · Codex-graded lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	6	8.528–8.579 (3D-NLTE)	fit window 0.06, fit / line-to-line scatter 0.04
red-optical	C I · Asplund et al. 2021 lines	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	10	8.488–8.536 (3D-NLTE)	fit / line-to-line scatter 0.04, model atmosphere 0.03

Excluded lines

These lines are left out of the Sun's measurement. They stay in the line lists: a cooler or more evolved star may measure them well. Blended lines are not excluded -- the spectrum synthesis models the blend. 1 lines.

Saturated: on the flat part of the curve of growth, so the line barely responds to the abundance (reduced width $\log W/\lambda$ above -4.9).

Species	Wavelength (Å, air)	Why it is excluded
C I	5380.340	saturated

Solar carbon context

Carbon is traced by atomic C I, including forbidden [C I], and molecular CH, C₂ and CO indicators. Each probes different line formation and molecular-equilibrium sensitivities. Indicator families and atmospheric treatments remain separate products.

[C/O indicator strategy](#) · [Methodology](#)

Download the evidence

[Download PDF](#) — Solar C appendix

[Source feed](#) — includes quarantined fit outputs, not adopted abundances

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned science snapshot. Source commit c730d5e72e3f8902d1d4d800b46fe89fbcafee69

Feed C.json v1.163 · updated 2026-10-03T22:11:25Z

Generated 2026-10-09T16:35:54-06:00 (source commit timestamp)

Feed SHA-256 f3146e51b5f7493e4bcec5843bb4cfbfab7995f56be16c85668a009ed80c23b5

Generator scripts/generate_cno_publication.py v1.0.0

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar C atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded C I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

Atomic data and gf authorities. AGSS21 names Li et al. for its C I gf lineage. This reference calculation is distinct from the campaign's NIST/MCHF adjudication; no gf source is substituted here. Sources: [5].

Model and grid sources. Registered C I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE / 1D-NLTE grids (CDS J/A+A/630/A104). The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [6].

Atomic data and gf authorities. Atomic C I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [7].

Molecular line data and delivery. C2 line-data delivery: C2 Swan $d^3\Pi_g - a^3\Pi_u$ (Brooke et al. 2013 / Masseron compilation). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [7], [8].

Molecular line data and delivery. CH line-data delivery: CH A-X: Masseron et al. 2014 (MNRAS 442, 1489) / 2022 update. Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpd.l.mpg.de, split per wavelength region) — see molecules-dir README.md The primary intake identifies Masseron 2014 as A&A 571 A47; the older delivery manifest's MNRAS locator conflicts with that record. Sources: [2], [7], [9].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 A) Sources: [6].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 A) Sources: [6].

Product-to-source index

VIS · HARPS · C I 5052 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-45eae148b981

VIS · HARPS · C I 5380 Å (single line) · 1D LTE · [1], [3], [4], [7] C-products-7718d3380b98

VIS · HARPS · C₂ Swan band (molecular band) · 1D LTE · [1], [2], [7], [8] C-products-8614c00a6f61

VIS · HARPS · CH G-band (molecular band) · 1D LTE · [1], [2], [7], [9] C-products-7eda3b79b64e

VIS · HARPS · C I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [7] C-products-475299546416

VIS · HARPS · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [6] C-products-d7a3b4042e96

VIS · HARPS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [6] C-products-f13deae5012f

VIS · HARPS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#) C-products-cff867aaaff3

VIS · HARPS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#) C-products-ccb0d66882f4

VIS · IAG FTS · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#) C-products-4382c8d47702

VIS · IAG FTS · C I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#) C-products-1296f26f4e5e

VIS · IAG FTS · C I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#) C-products-5c2d1868fca6

VIS · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)
C-products-750df87348c4

VIS · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
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C-products-897540c2aad7

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red-optical · IAG FTS · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
C-products-8fb53bf46822

red-optical · IAG FTS · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#) C-products-246b5509c29e

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C-products-eb9499c9a281

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
C-products-7304be6c64c8

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
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red-optical · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)
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red-optical · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 3D NLTE (Amarsi) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
C-products-9f874a36d2c8

red-optical · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D NLTE (Bergemann) · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[6\]](#)
C-products-e70e17e756dc

VIS · HARPS · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#) C-quarantine-450b744f3f97

VIS · Kitt Peak (Kurucz 2005) · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#) C-quarantine-2f69a108b1e7

VIS · Kitt Peak (1984, molecfit) · C I · Codex-graded lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#) C-quarantine-4e2a7411ee02

red-optical · Kitt Peak (Kurucz 2005) · C I · Asplund et al. 2021 lines · 1D LTE · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[7\]](#)
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